CURRICULUM VITAE

Aman Dimri
Department Of Chemistry, Stony Brook, New York USA, 11790
Email ✉ : aman.dimri@stonybrook.edu
Phone 🔊: +1 6318205562

(a) Education & Training:

Ph.D. Student — CGPA-3.91/4
Department Of Chemistry, Stony Brook, NY, USA — 2021-Current
Field Of Study: Physics

Integrated MSc. — CGPA-9.39/10
School Of Physical Sciences, NISER, Jatani, Bhubaneshwar, India — 2016-2021
Field Of Study: Physics

Higher Secondary School Education (10+2) — 95.6%
Doon International School, Dehradun, Uttarakhand, India — 2015

Secondary School Education (10th Boards) — Grade Point-10/10
Doon International School, Dehradun, Uttarakhand, India — 2013

(b) Scholastic Achievements & Fellowships

- Qualified for Prelims Of National Anveshika Experiment Skill Test 2020, and was awarded for outstanding performance in the screening test.
- Summer Fellow Of The Indian Academy Of Sciences (IAS- Summer Research Program), 2019. IAS-Summer Research Fellowship is awarded to bright students to work with scientists associated with the Academies on research-oriented projects. Out of thousands of applications in 2019, only 188 were selected for this prestigious fellowship in the physics department (IAS Fellowship 2019).
- Recipient Of The Academic Excellence Award, School Of Physical Sciences, NISER in the academic years, 2016-2019.
- KVPY Fellow (Kishore Vaigyanik Protsahan Yojana, Funded by DST, Government Of India), 2017-2021. The "Kishore Vaigyanik Protsahan Yojana" (KVPY) is a program started in 1999 by the Department of Science and Technology (DST), Government of India to encourage students who are studying Basic Sciences to take up research career in Science. The aim of the program is to identify and encourage talented and motivated students to pursue career in research (KVPY). In 2016, a total of 97,467 students were shortlisted to appear for KVPY exam and 2,531 students were awarded the KVPY scholarship across all eligible streams.
- DST-Inspire Fellow (Scholarship for Higher Education (SHE) under Innovation in Science Pursuit for Inspired Research (INSPIRE) by Department Of Science And Technology (DST), Gov-

ernment Of India), 2016-2017. INSPIRE is an innovative program sponsored by the DST for attraction of talent to Science (Inspire Fellowship). I was awarded this eminent scholarship for securing a rank of 84 out of 50,000 applicants in the National Entrance Screening Test (NEST).

(c) Internships & Projects:

- Masters' Thesis: Particle Production In High Energy Collisions 2020-2021
  Guided By- Professor Bedangadas Mohanty, School Of Physical Sciences, NISER, Bhubaneshwar

  In this project, I employed the Hadron Resonance Gas Model to gain a comprehensive understanding of the intricate system generated during high-energy collisions of heavy ions. This is combed with AMPT and hydro-models to delve into the particle production mechanism within the hot and dense medium formed during the collisions..

- Open-Ended Lab Projects:
  - Building A Scintillator Detector: 2020
    Guided By: Dr Subhankar Bedanta and Dr Varchaswi KS Kashyap, School Of Physical Sciences, NISER, Bhubaneshwar

    The primary objective of this project is to gain a comprehensive understanding of the underlying principles governing scintillator detectors, followed by the practical implementation of these principles to construct a highly functional plastic scintillation detector.

  - Setting Up Diode Pumped Nd:YAG Laser: 2019
    Guided By: Dr Ashok K Mohapatra and Dr Gunda Santosh Babu, School Of Physical Sciences, NISER, Bhubaneshwar

    The aim of this project is to delve into the fundamental principles underlying the operation of lasers, with a specific focus on the Nd-YAG Laser, which belongs to the category of Solid State Lasers. By acquiring an in-depth understanding of these principles, the goal is to successfully construct and demonstrate the functionality of an Nd-YAG Laser

  - The Faraday Effect: 2019
    Guided By: Dr Ashok K Mohapatra and Dr Gunda Santosh Babu, School Of Physical Sciences, NISER, Bhubaneshwar

    The fundamental purpose of this experiment is to observe and comprehend the intriguing phenomenon of Faraday rotation. This phenomenon involves the rotation of the plane of polarization of light as it interacts with an axial magnetic field within a medium, resulting in a measurable angle of rotation. By conducting this experiment, our aim is to gain valuable insights into the intricate dynamics of Faraday rotation and its implications within the realm of optics.

- An Elementary Insight Into Quark Model And Quantum Chromodynamics 2019
  Guided By: Professor Anirban Kundu, University Of Calcutta, Kolkata

  This project focuses on understanding the formation of the quark model as a direct consequence of symmetry of the fundamental particles over which the nuclear interaction does not differentiate.

- The Relativistic Boltzmann Equation 2019
  Guided By: Dr Amaresh Kumar Jaiswal, School Of Physical Sciences, NISER, Bhubaneshwar

  In this project, I explored the fundamental concepts of relativistic Kinetic theory. Specifically, I derived the relativistic Boltzmann equation, which governs the evolution of the distribution function across phase space and time.

- Numerical Analysis Of Linear, Non-Linear And Ordinary Differential Equations 2018
  Guided By- Professor Bedangadas Mohanty, School Of Physical Sciences, NISER, Bhubaneshwar

  In the field of Physics, complex systems of equations often surpass manual analytical solving and necessitate computational power. This project delves into elementary algorithms employed for solving various equation systems, encompassing linear, nonlinear, and ordinary differential equations. By examining their time complexity and solution accuracy, a comparative analysis is conducted to evaluate the performance and reliability of these algorithms. The aim is to leverage computational capabilities to enhance problem-solving efficiency in Physics.

(d) Conferences Attended:

- Presented a poster in QM-2025 titled: "Resolving the $C_4\{4\}$ Sign Change Puzzle in Pb-Pb Collisions"
- Presented a poster in QM-2023 titled: "Effect of Nuclear Shape Parameters On Initial State Observables In Heavy Ion Collisions"
- Presented a talk tilted "Nuclear shape fluctuations in heavy ion collisions" at INT-23-1a: Intersection of nuclear structure and high-energy nuclear collisions in 2023, showcasing a consistent way to parameterize the effects of shape fluctuations on initial stages of heavy ion collisions.
- Contributed an oral presentation to Fall Meeting for APS DNP-2022 titled "Probing Effect of Nuclear Shape Coexistence in Heavy-Ion Collisions using Glauber Model".
- Presented a poster in SQM 2022: The 20th International Conference on Strangeness in Quark Matter.
- Attended Symposium on Heavy-Ion Physics at FAIR, RHIC and LHC Facilities, NISER, Bhubaneshwar, 2018. The aim of this two day symposium was to introduce it's participants to the current research and future work in heavy-ion physics at various aforementioned facilities.
- Attended National Science Vijyoshi Camp held at Kolkata, India, 2016 & 2017. The aim of the

annual National Science Vijyoshi Camp is to provide a forum for interactions between bright young students and leading researchers in various branches of science and mathematics. Around 1200 students participate in this renowned science camp (Vijyoshi 2016 & Vijyoshi 2017).

(e) Publications:

- Impact of nuclear shape fluctuations in high-energy heavy ion collisions :
  We investigate the impact of nuclear shape fluctuations on initial state geometry in heavy ion collisions, focusing on eccentricity (ε_2) and inverse transverse size ($d_\perp$). Our aim is to quantify the effects of these parameters on initial state observables in a coherent manner. We show that overall quadrupole deformation fluctuations enhance the cumulants of ε_2 and $d_\perp$, while triaxiality fluctuations reduce the differences between prolate and oblate nuclei. Our results suggest that, in the large fluctuation limit, the initial state observables' values approach those obtained in collisions of rigid triaxial nuclei.
- Evidence for the collective nature of radial flow in Pb+Pb collisions with the ATLAS detector:
  Anisotropic and radial flow are essential probes of quark-gluon plasma (QGP) dynamics, but radial flow has been less explored experimentally. This study presents the first measurement of p_T differential radial flow fluctuations in Pb+Pb collisions at $\sqrt{s_{NN}}$ = 5.02 TeV. The results show clear signatures of collective radial flow, including long-range pseudorapidity correlations, pT factorization, and consistent shapes across centralities. Comparison with hydrodynamic models highlights radial flow's sensitivity to QGP bulk viscosity, providing a new tool to investigate the plasma's collective behavior and transport properties.*
- Disentangling the global multiplicity and spectral shape fluctuations in radial flow:
  Radial flow is a key collective phenomenon in heavy-ion collisions, manifests through event-by-event fluctuations in transverse momentum (p_T) spectra. The p_T-differential radial flow, $v_0(p_T$, initially conceived to capture local spectral shape fluctuations, is influenced by global multiplicity fluctuations. Using the HIJING model, we explore how different definitions of event activity for centrality and spectral normalization schemes affect $v_0(p_T$. We find these methodological variations induce a constant offset in $v_0(p_T)$ without altering its shape, indicating that the dynamic p_T-differential information on radial flow remains robust, but its absolute magnitude is meaningful only up to a baseline offset dictated by global multiplicity fluctuations.*

*Contributing member of the Primary Analysis Group